

THE PERSISTENT Π_1 & THE LOWER CENTRAL QUOTIENTS BARCODES

GONZALO ORTEGA CARPINTERO

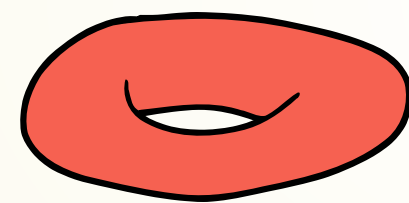
gonzalo.ortega-carpintero@city.ac.uk



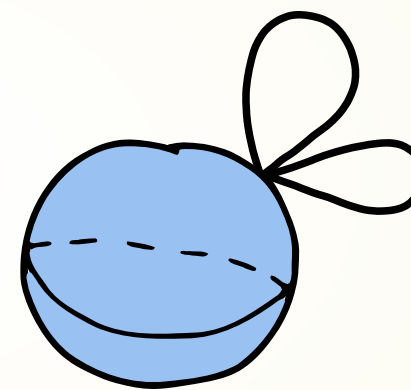
A joint work with Ximena Fernández at

CITY
ST GEORGE'S
UNIVERSITY OF LONDON

THE LIMITATIONS OF H_*

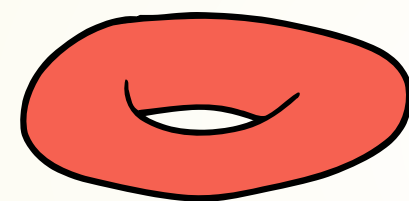


$$T = S_1 \times S_1$$

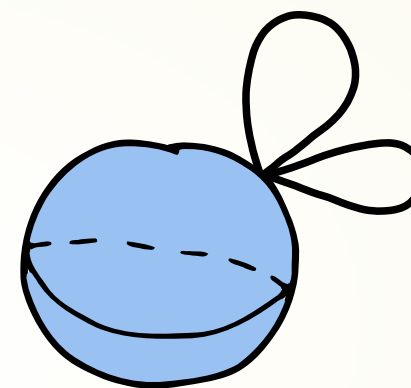


$$X = S^2 \vee S^1 \vee S^1$$

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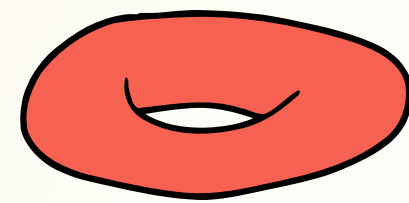
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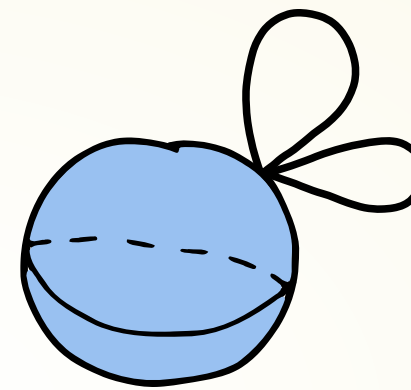
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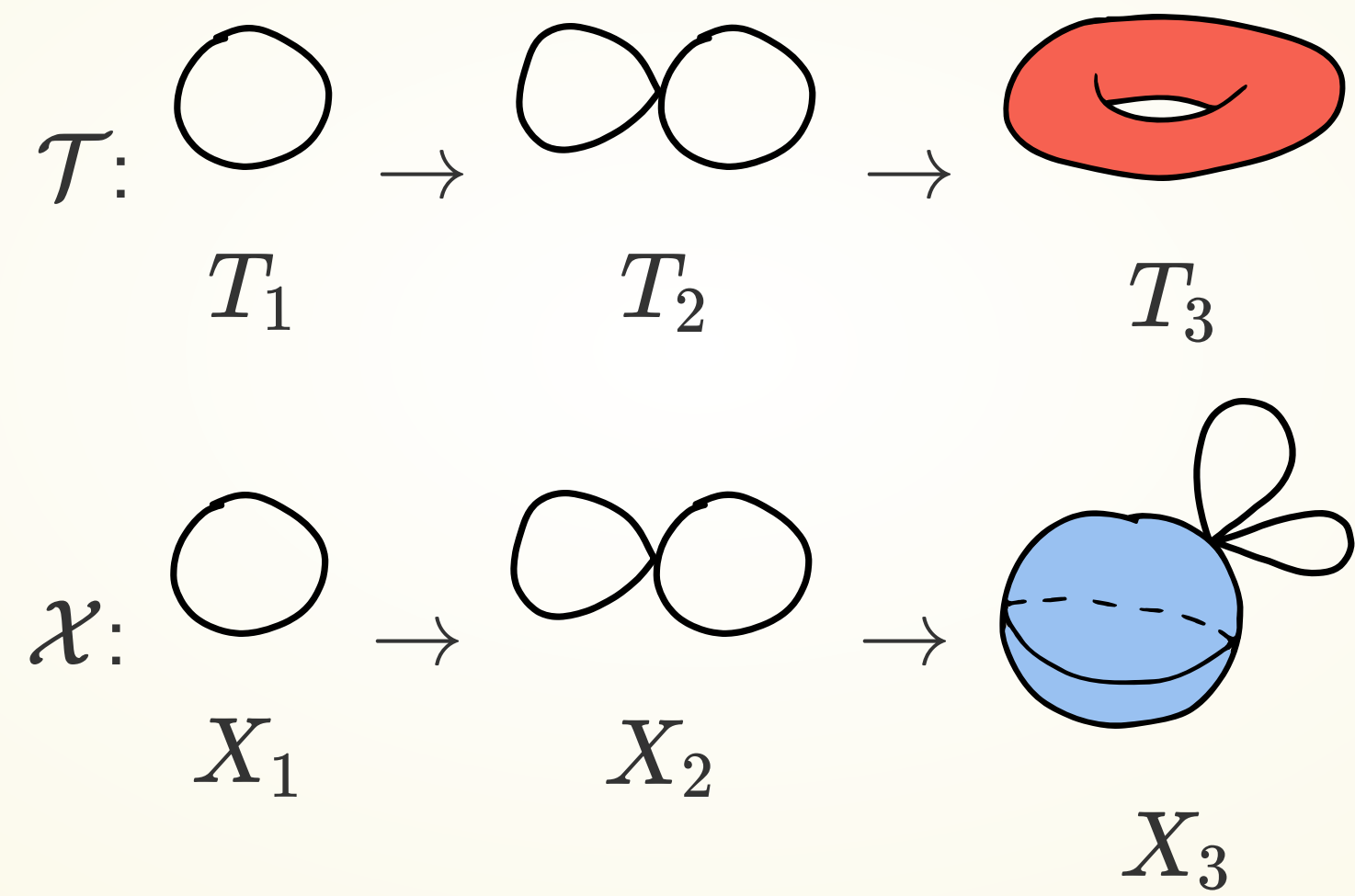


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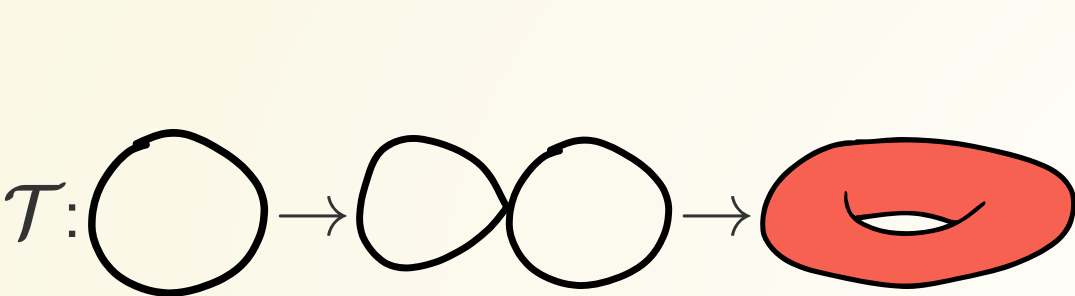
$$H_1(T) = \mathbb{Z} \oplus \mathbb{Z} = H_1(X)$$

$$\pi_1(T) = \mathbb{Z} \oplus \mathbb{Z} \neq \mathbb{Z} * \mathbb{Z} = \pi_1(X)$$

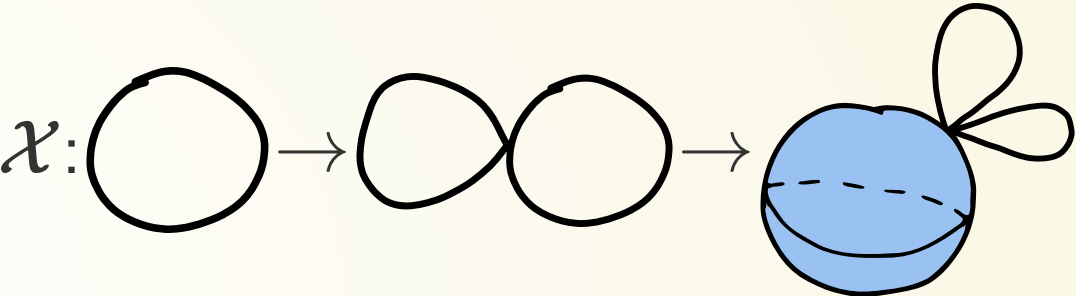
THE LIMITATIONS OF PERSISTENT H_*



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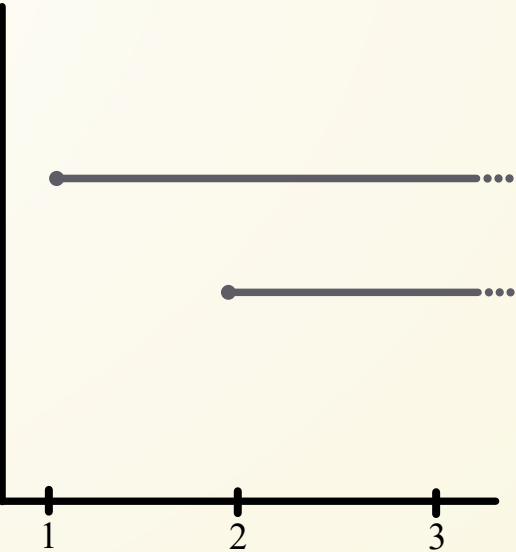
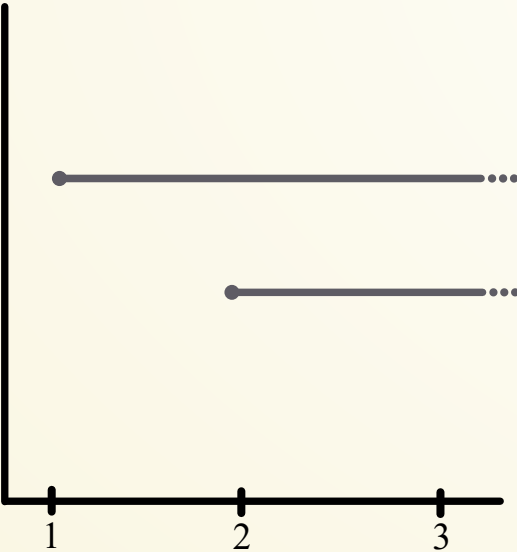
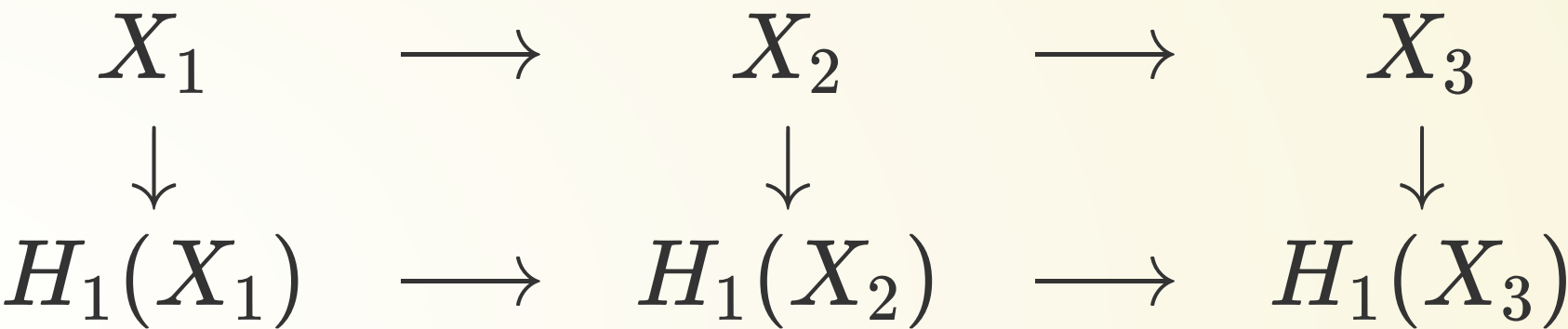
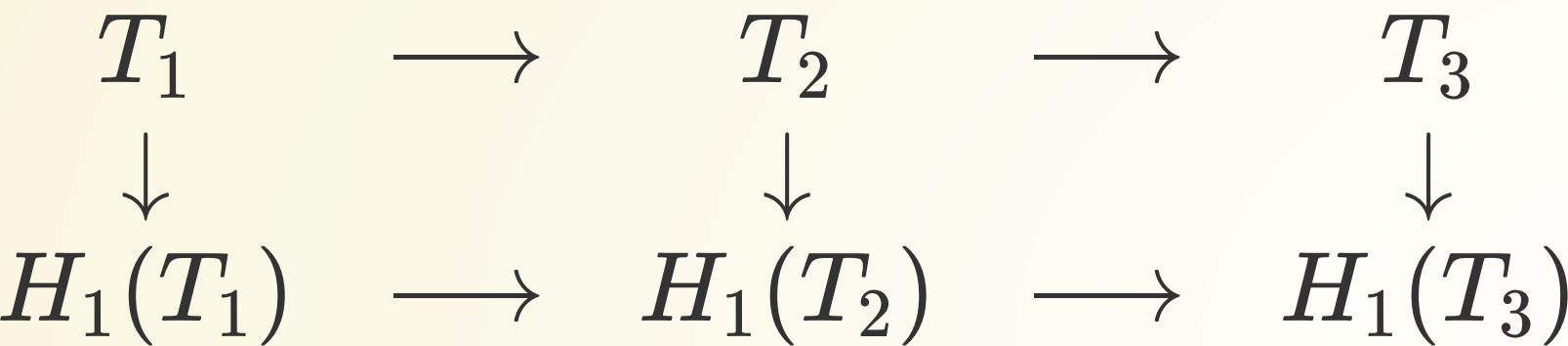
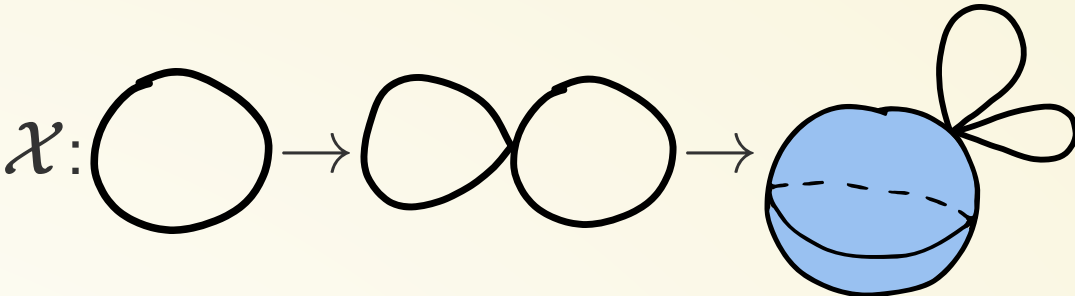
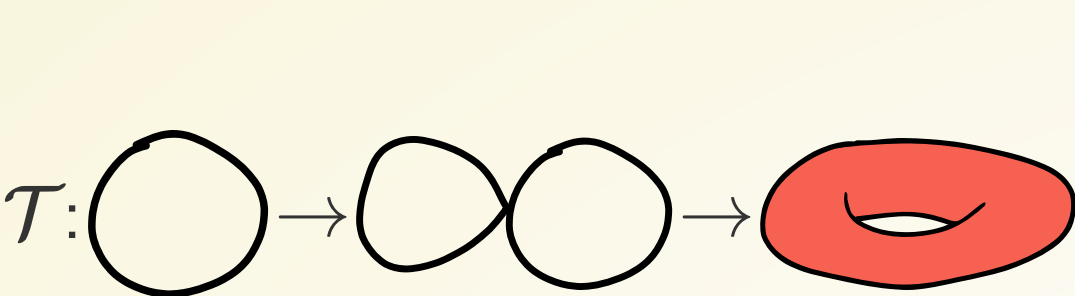


T_1	$\longrightarrow$	T_2	$\longrightarrow$	T_3
$\downarrow$		$\downarrow$		$\downarrow$
$H_1(T_1)$	$\longrightarrow$	$H_1(T_2)$	$\longrightarrow$	$H_1(T_3)$



X_1	$\longrightarrow$	X_2	$\longrightarrow$	X_3
$\downarrow$		$\downarrow$		$\downarrow$
$H_1(X_1)$	$\longrightarrow$	$H_1(X_2)$	$\longrightarrow$	$H_1(X_3)$

THE LIMITATIONS OF PERSISTENT H_*



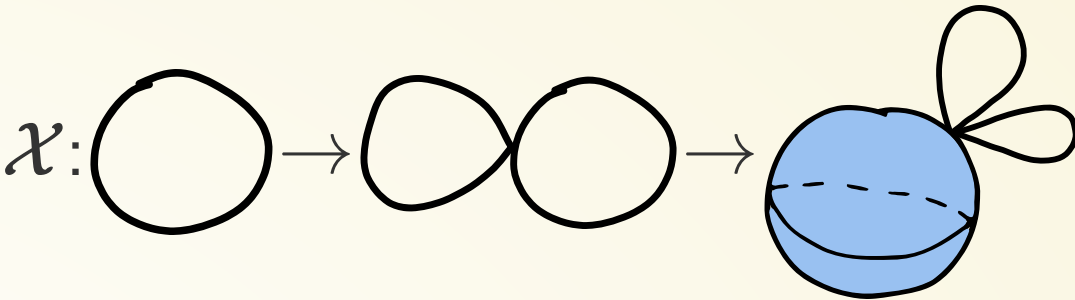
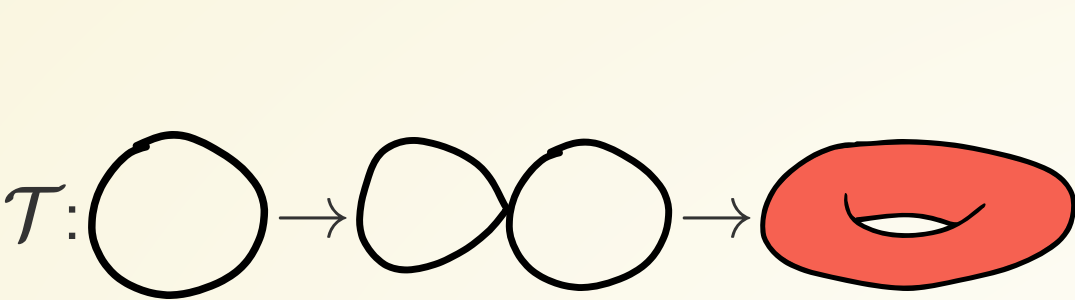
$PH_1(\mathcal{T}) = PH_1(\mathcal{X})$

THE LIMITATIONS OF PERSISTENT H_*



WHAT WOULD WE LIKE TO DO?

THE LIMITATIONS OF PERSISTENT H_*



$$\begin{array}{ccccc} T_1 & \longrightarrow & T_2 & \longrightarrow & T_3 \\ \downarrow & & \downarrow & & \downarrow \\ H_1(T_1) & \longrightarrow & H_1(T_2) & \longrightarrow & H_1(T_3) \end{array}$$

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Theorem (Forman, 1998): Suppose K is a regular CW complex with a discrete Morse function. Then K is homotopy equivalent to a CW complex with exactly one cell of dimension p for each critical simplex of dimension p .

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Theorem (Fernández, 2024): A reduced presentation of $\pi_1(K)$ can be computed with complexity $\mathcal{O}(N^2)$, where N depends on the total number of cells in K .

CAN Π_1 BE COMPUTED? YES*

CAN Π_1 BE COMPUTED OVER A FILTRATION?

CAN Π_1 BE COMPUTED? YES*

CAN PERSISTENT Π_1 BE COMPUTED?

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$$\begin{array}{ccccccc} K_1 & \longrightarrow & K_2 & \longrightarrow & \cdots & \longrightarrow & K_n \\ \downarrow & & \downarrow & & & & \downarrow \\ \pi_1(K_1) & \longrightarrow & \pi_1(K_2) & \longrightarrow & \cdots & \longrightarrow & \pi_1(K_n) \end{array}$$

Naive computation: Compute π_1 presentations independently over every element in the filtration.

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Efficient computation: Make an single computation through all the filtration.

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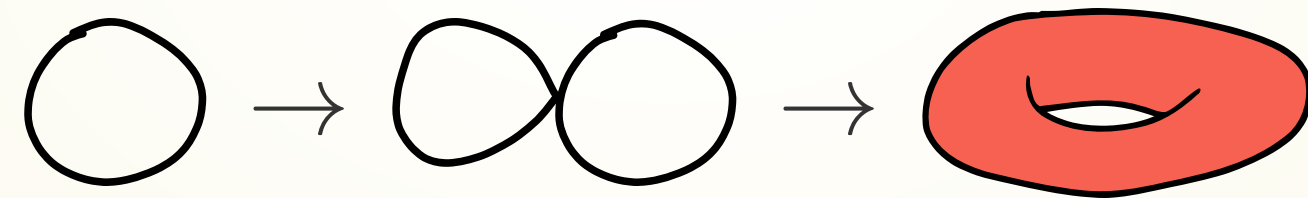
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Efficient computation: Make an single computation through all the filtration.

Theorem: *There exists an algorithm of complexity $\mathcal{O}(N^2)$ that computes presentations of all $\pi_1(K_\alpha)$ for all $1 \leq \alpha \leq n$, where N depends on the total number of cells in K_n .*

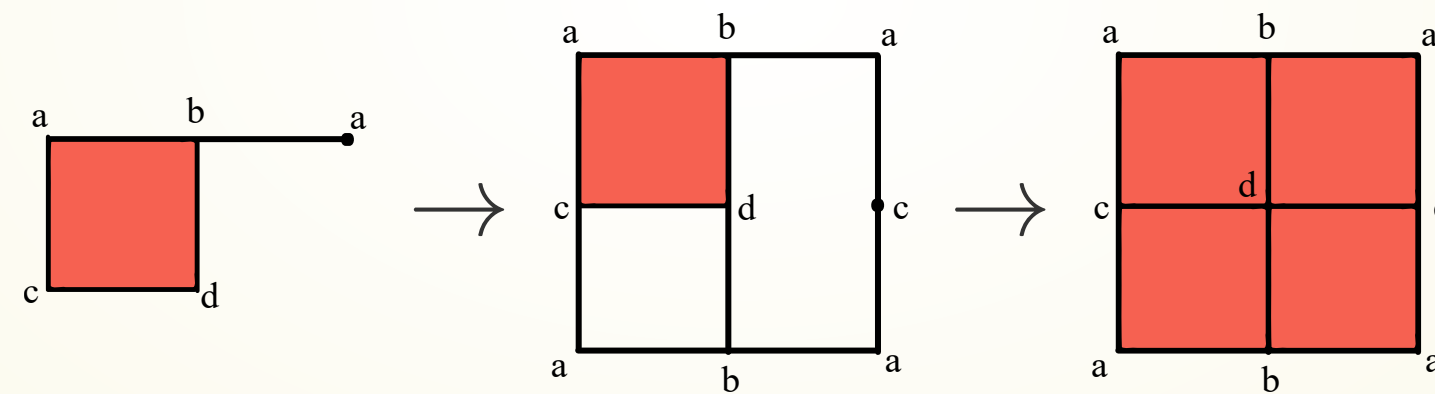
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The groups $\pi_1(K_\alpha)$ in

$$\pi_1(K_1) \longrightarrow \pi_1(K_2) \longrightarrow \cdots \longrightarrow \pi_1(K_n)$$

may **NOT** be abelian!

CAN Π_1 BE COMPUTED? **YES***

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The groups $\pi_1(K_\alpha)$ in

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may **NOT** be abelian!

Also, we only know presentations of each $\pi_1(K_\alpha)$, but **NOT** the actual group!

LOWER CENTRAL QUOTIENTS

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Definition (Lower central series): The **lower central series** of a group G is the descending series of normal subgroups

$$G = \gamma_1(G) \supseteq \gamma_2(G) \supseteq \cdots \supseteq \gamma_n(G) \supseteq \cdots$$

where

$$\gamma_{n+1}(G) = [\gamma_n(G), G] = \langle [h, g] : h \in \gamma_n(G), g \in G \rangle.$$

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Definition (Lower central quotient): We define the **lower central quotient** of G as

$$lcq_n(G) := \frac{\gamma_n(G)}{\gamma_{n+1}(G)}.$$

LOWER CENTRAL QUOTIENTS

Let $G = \pi_1(K)$

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...

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For every n , $lcq_n(G)$ *IS ABELIAN!*

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For every n , $lcq_n(G)$ CAN BE COMPUTED! (Sims, 1994)

LOWER CENTRAL QUOTIENTS

*For every n , $lcq_n(G)$ **IS ABELIAN!***

*For every n , $lcq_n(G)$ **CAN BE COMPUTED! (Sims, 1994)***

*The **complexity** of computing $lcq_n(G)$ depends on the **number of generators and relations** of the presentation of G we are using.*

LOWER CENTRAL QUOTIENTS

$$\begin{array}{ccccccc}
 K_1 & \longrightarrow & K_2 & \longrightarrow & \cdots & \longrightarrow & K_n \\
 \downarrow & & \downarrow & & & & \downarrow \\
 \pi_1(K_1) & \longrightarrow & \pi_1(K_2) & \longrightarrow & \cdots & \longrightarrow & \pi_1(K_n) \\
 \downarrow & & \downarrow & & & & \downarrow \\
 lcq_n(\pi_1(K_1)) & \longrightarrow & lcq_n(\pi_1(K_2)) & \longrightarrow & \cdots & \longrightarrow & lcq_n(\pi_1(K_n))
 \end{array}$$

LOWER CENTRAL QUOTIENTS

Every $lcq_n(\pi_1(K_\alpha))$ is abelian so

$$lcq_n(\pi_1(K_1)) \longrightarrow lcq_n(\pi_1(K_2)) \longrightarrow \cdots \longrightarrow lcq_n(\pi_1(K_n))$$

can be seen as a graded abelian module.

LOWER CENTRAL QUOTIENTS

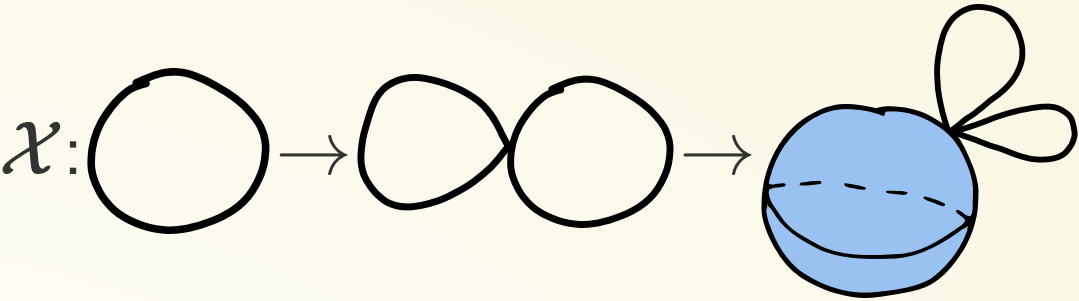
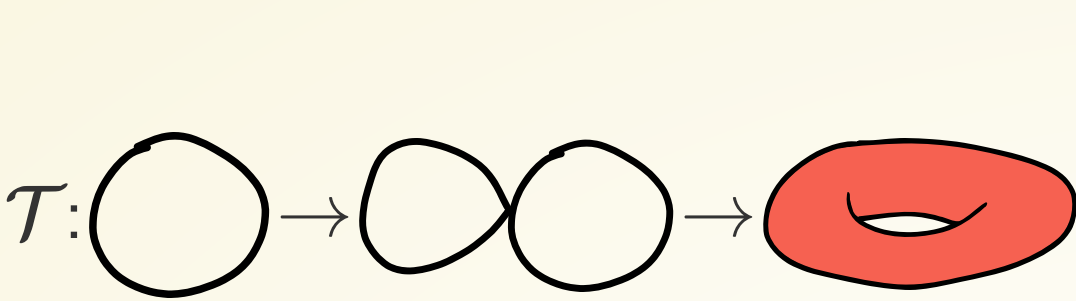
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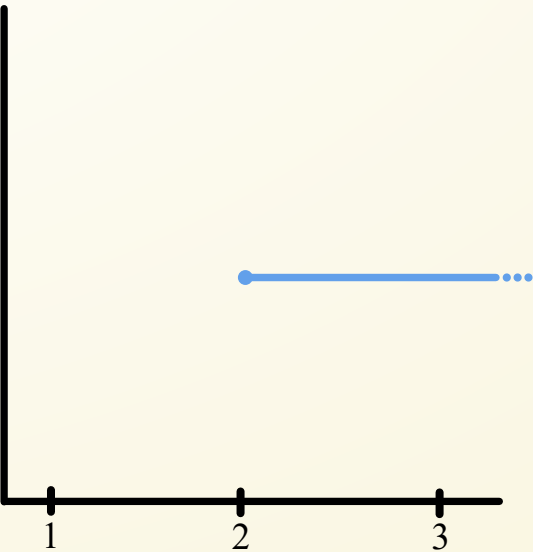
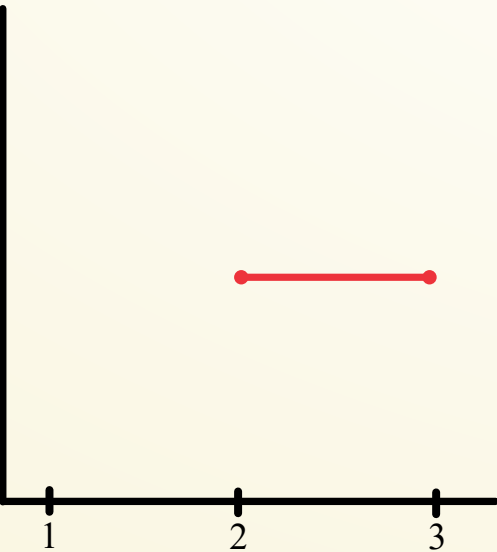
*We can compute **BARCODES!** (Skraba & Vejdemo, 2013)*

LOWER CENTRAL QUOTIENTS



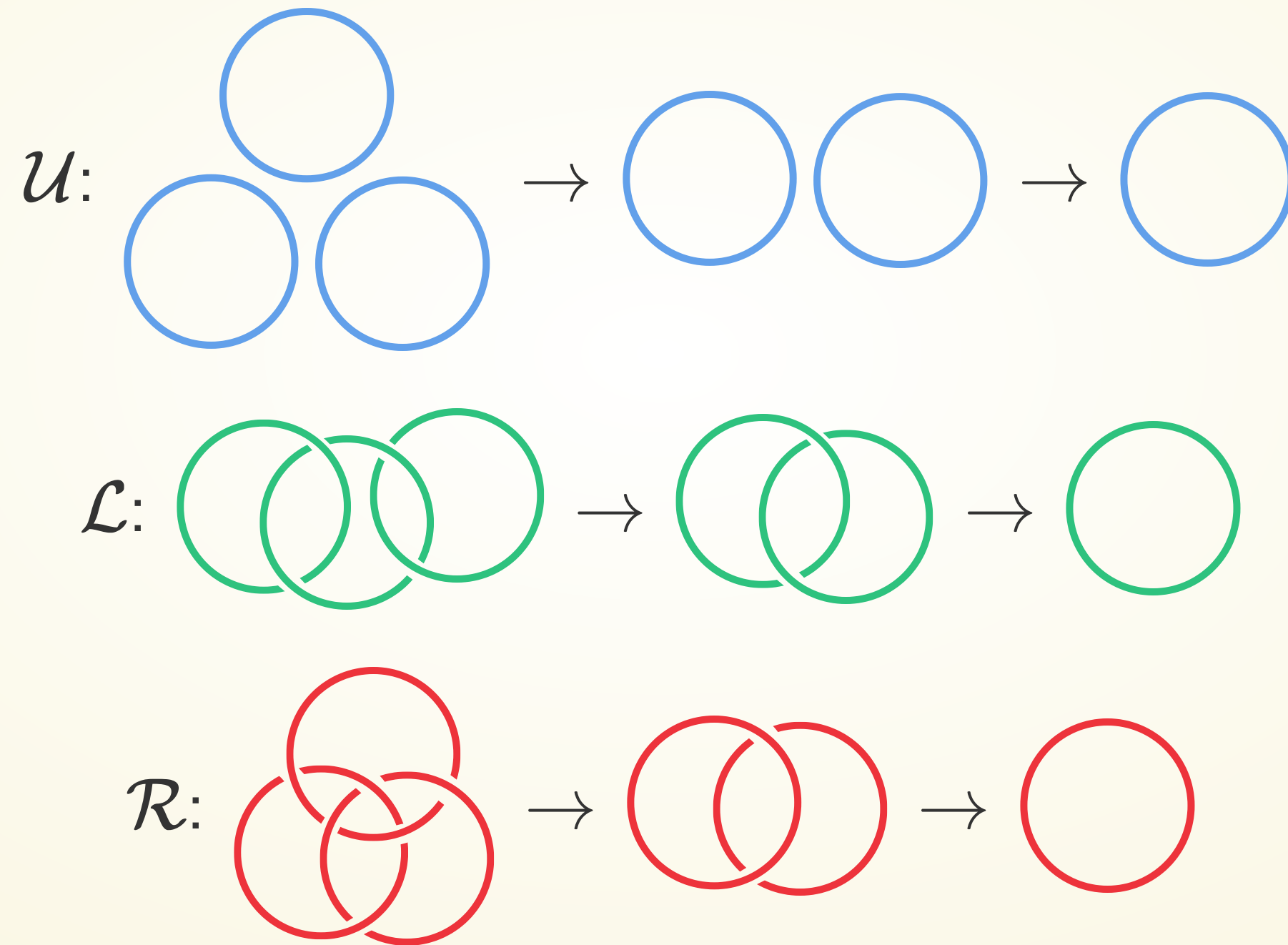
T_1	$\longrightarrow$	T_2	$\longrightarrow$	T_3
$\downarrow \pi_1$		$\downarrow \pi_1$		$\downarrow \pi_1$
$\mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} * \mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} \oplus \mathbb{Z}$
$\downarrow lcq_2$		$\downarrow lcq_2$		$\downarrow lcq_2$
0	$\longrightarrow$	$\mathbb{Z}$	$\longrightarrow$	0

X_1	$\longrightarrow$	X_2	$\longrightarrow$	X_3
$\downarrow \pi_1$		$\downarrow \pi_1$		$\downarrow \pi_1$
$\mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} * \mathbb{Z}$	$\longrightarrow$	$\mathbb{Z} * \mathbb{Z}$
$\downarrow lcq_2$		$\downarrow lcq_2$		$\downarrow lcq_2$
0	$\longrightarrow$	$\mathbb{Z}$	$\longrightarrow$	$\mathbb{Z}$

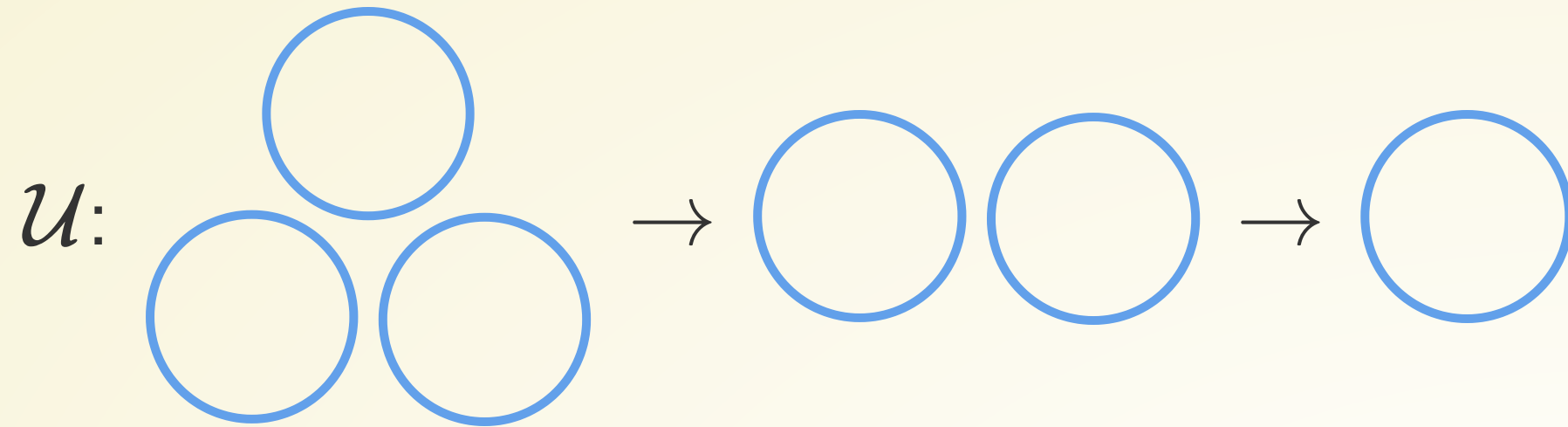


LOWER CENTRAL QUOTIENTS

Example: lcq_n barcodes of links.



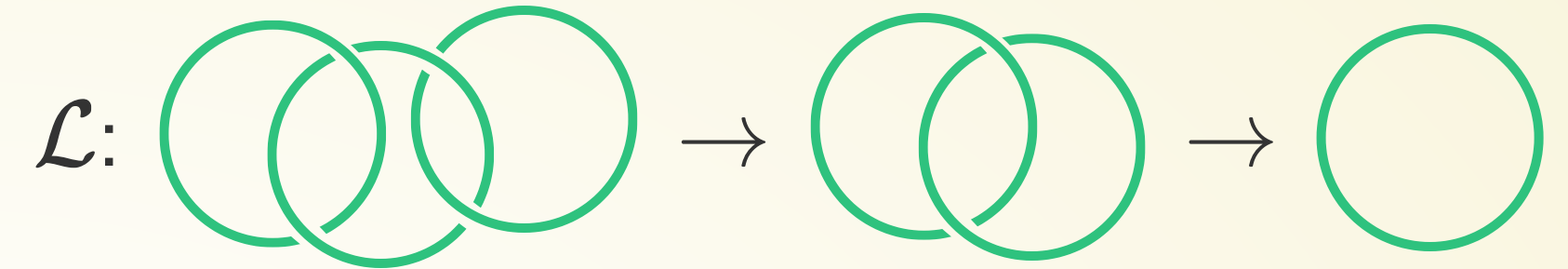
Example: lcq_n barcodes of links. (π_1)



$$\pi_1(\mathbb{R}^3 \setminus U_1) = \langle u_1, u_2, u_3 \mid - \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus U_2) = \langle u_1, u_2 \mid - \rangle$$

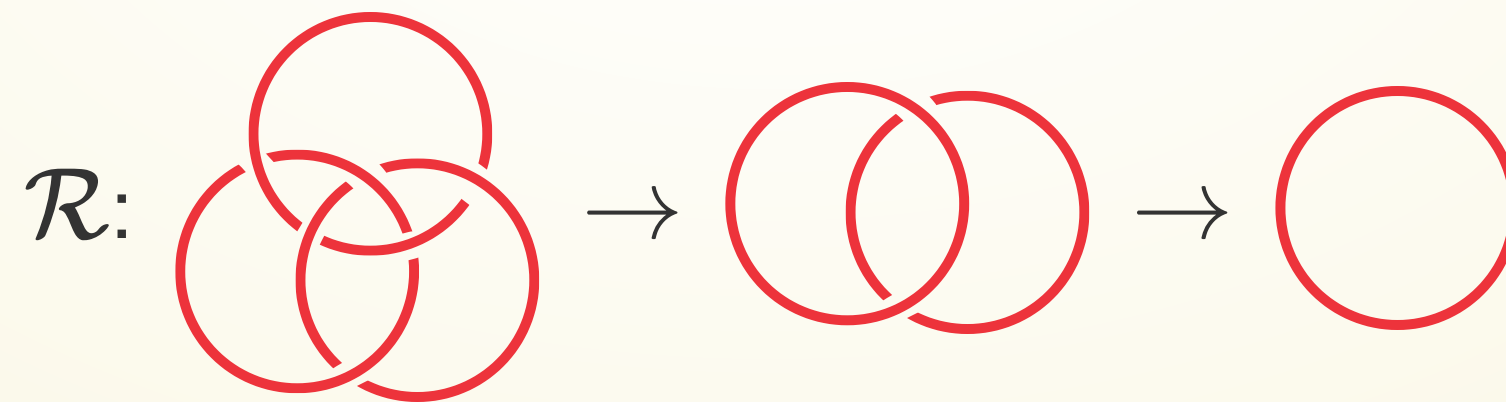
$$\pi_1(\mathbb{R}^3 \setminus U_3) = \langle u_1 \mid - \rangle$$



$$\pi_1(\mathbb{R}^3 \setminus L_1) = \langle l_1, l_2, l_3 \mid [l_1, l_2], [(l_1 l_3), l_2] \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus L_2) = \langle l_1, l_2 \mid [l_1, l_2] \rangle$$

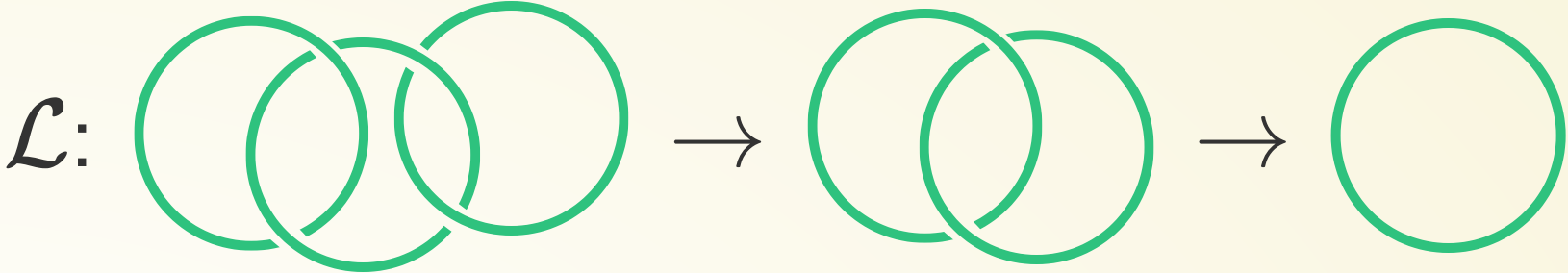
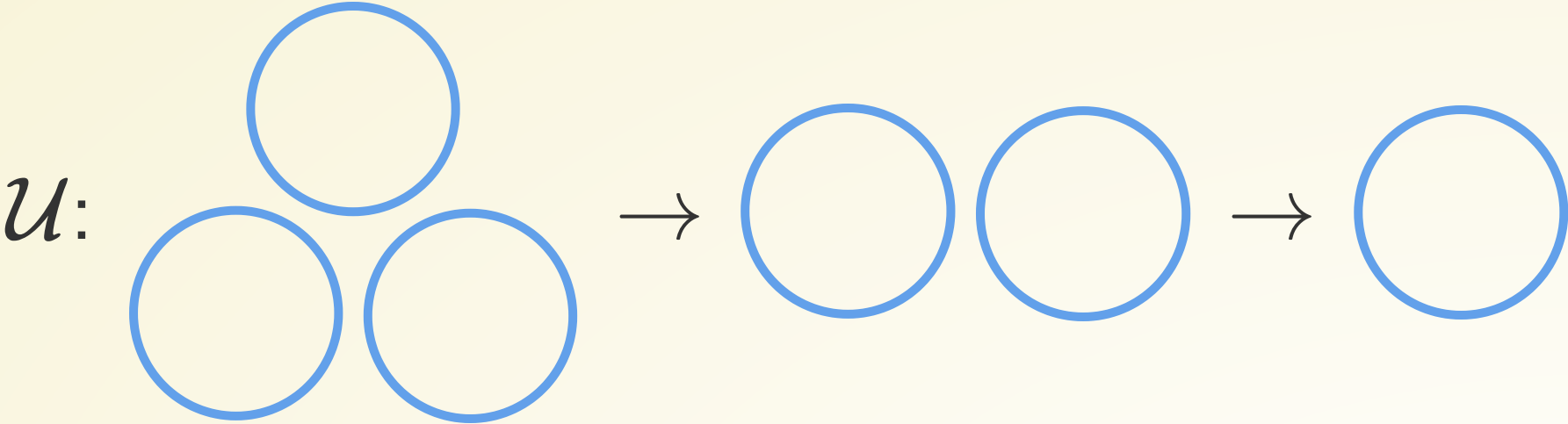
$$\pi_1(\mathbb{R}^3 \setminus L_3) = \langle l_1 \mid - \rangle$$



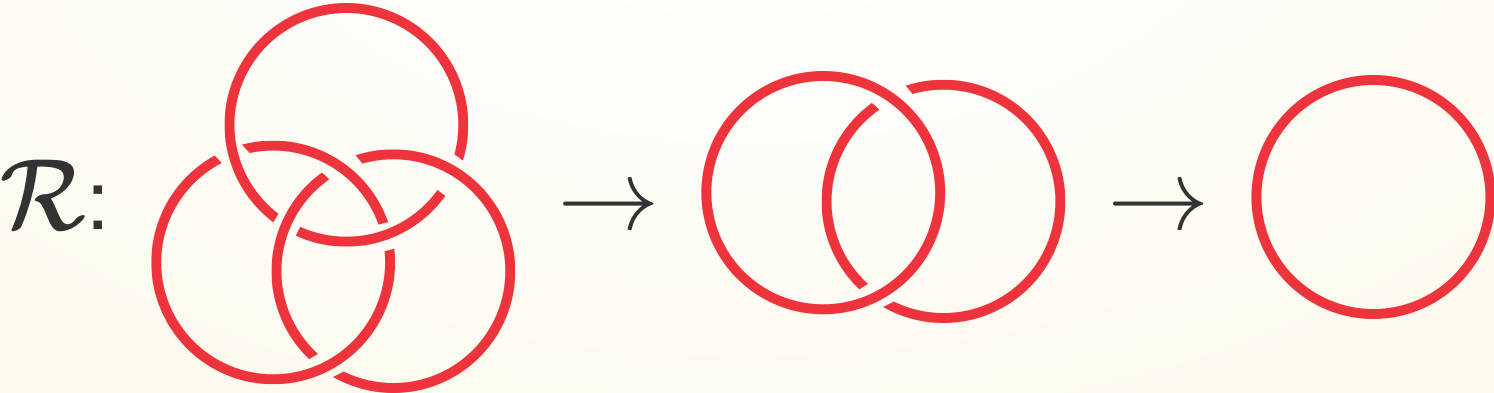
$$\pi_1(\mathbb{R}^3 \setminus R_1) = \langle r_1, r_2, r_3 \mid r_3 r_2^{-1} r_3^{-1} r_1^{-1} r_3 r_2 r_3^{-1} r_2^{-1} r_1 r_2, r_3 r_2 r_3^{-1} r_1 r_3 r_1^{-1} r_2^{-1} r_1 r_3^{-1} r_1^{-1} \rangle$$

$$\pi_1(\mathbb{R}^3 \setminus R_2) = \langle r_1, r_2 \mid - \rangle \quad \pi_1(\mathbb{R}^3 \setminus R_3) = \langle r_1 \mid - \rangle$$

Example: lcq_n barcodes of links. ($lcq_1 = H_1$)

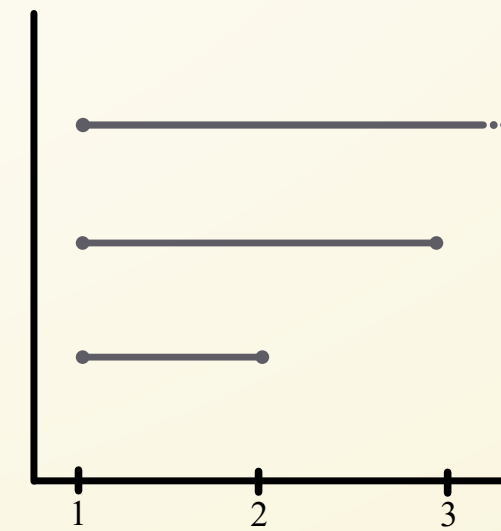
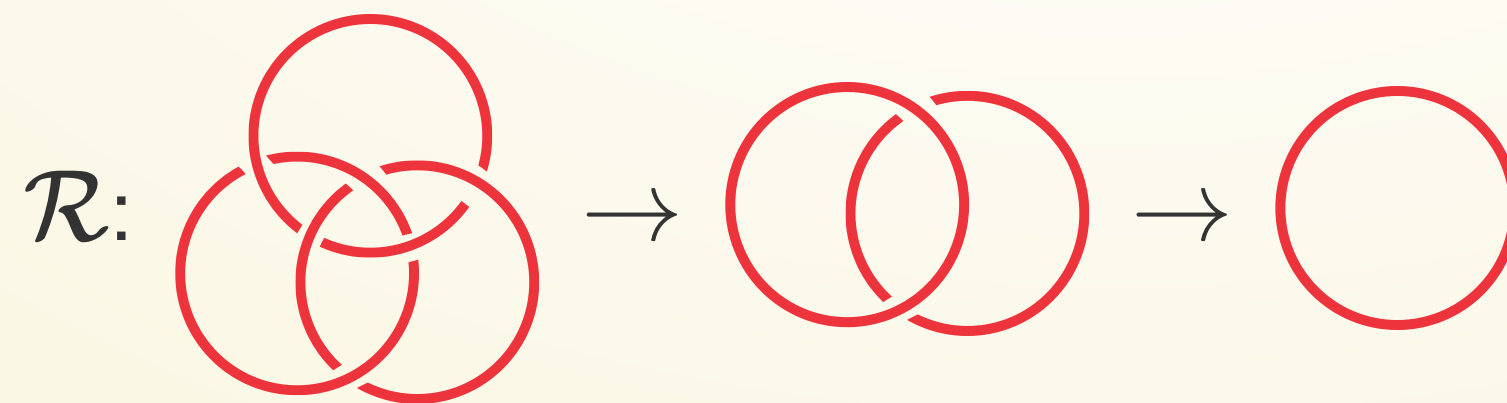
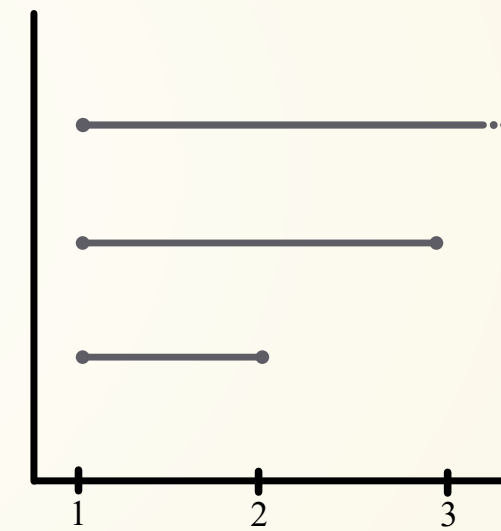
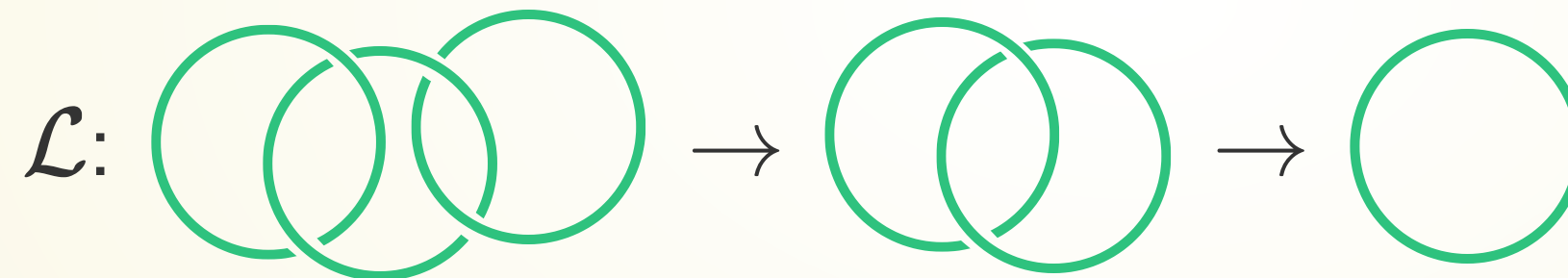
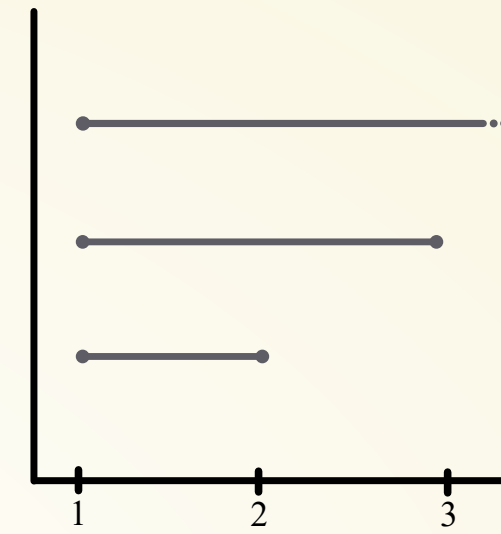
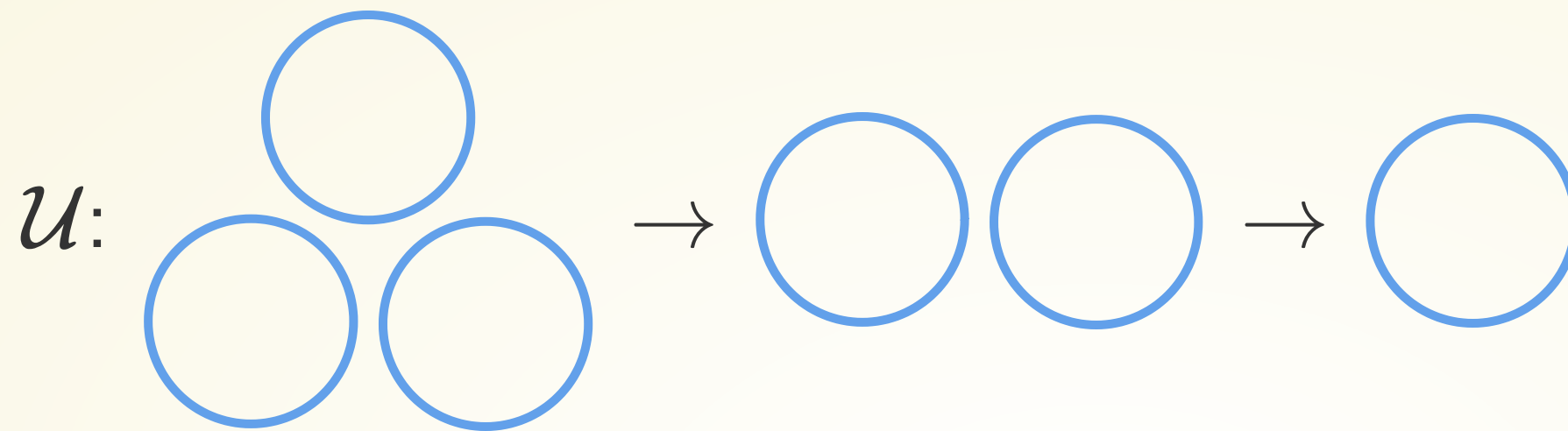


$\mathbb{R}^3 \setminus U_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_3$	$\mathbb{R}^3 \setminus L_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_3$
$\downarrow H_1$		$\downarrow H_1$		$\downarrow H_1$	$\downarrow H_1$		$\downarrow H_1$		$\downarrow H_1$
$\mathbb{Z}^3$	$\longrightarrow$	$\mathbb{Z}^2$	$\longrightarrow$	$\mathbb{Z}$	$\mathbb{Z}^3$	$\longrightarrow$	$\mathbb{Z}^2$	$\longrightarrow$	$\mathbb{Z}$

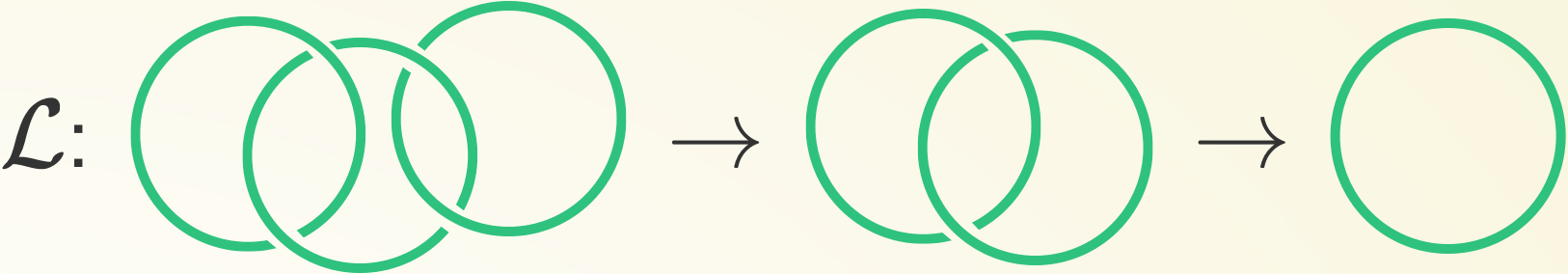
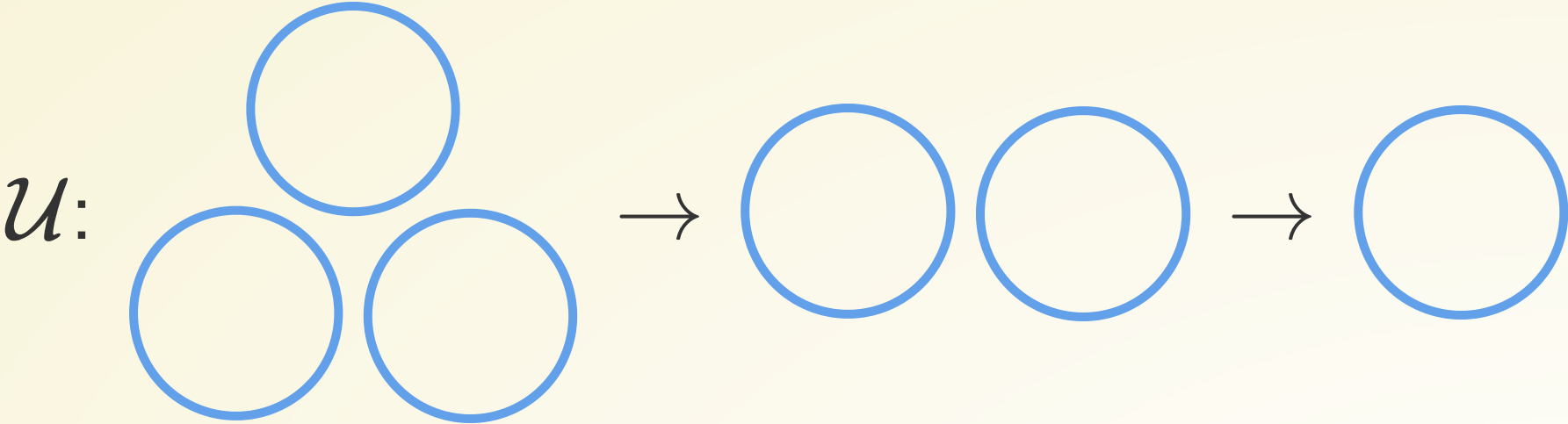


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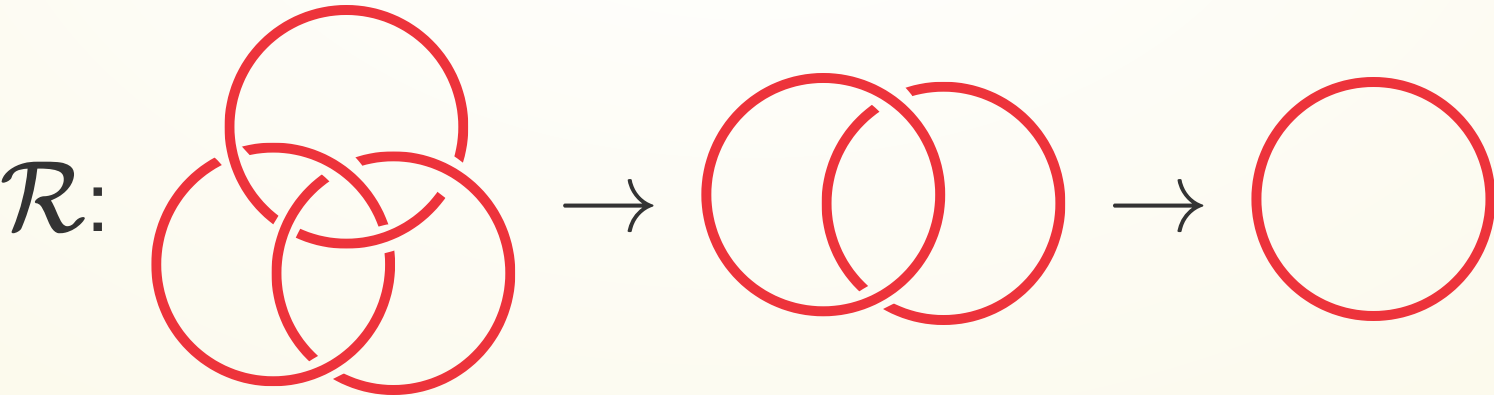
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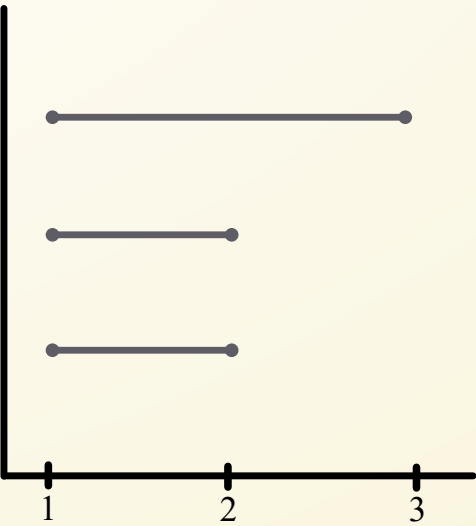
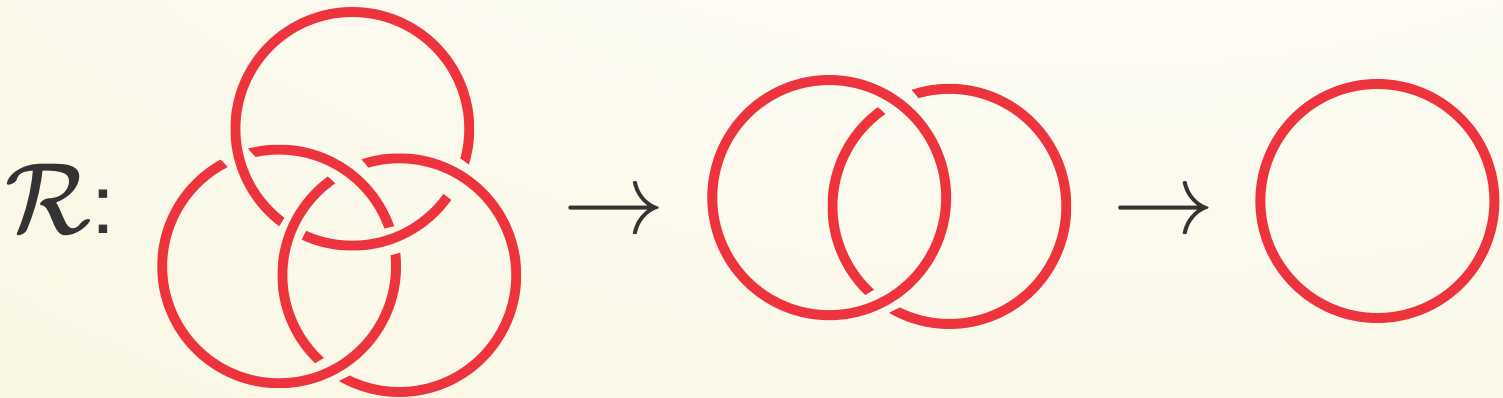
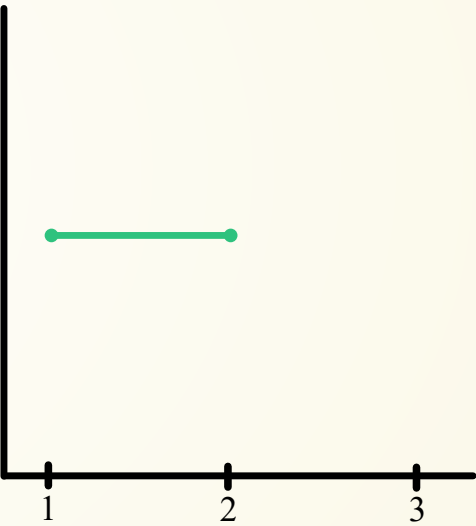
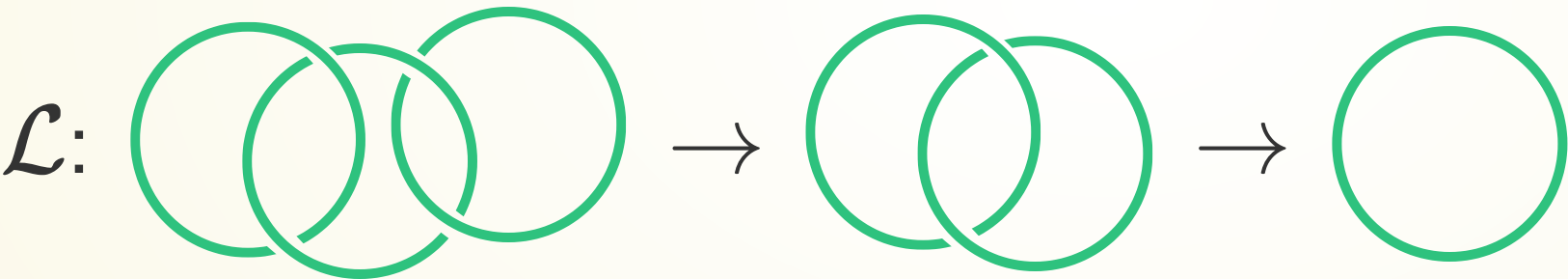
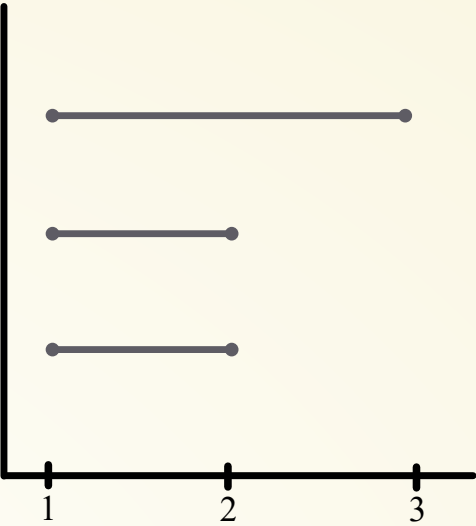
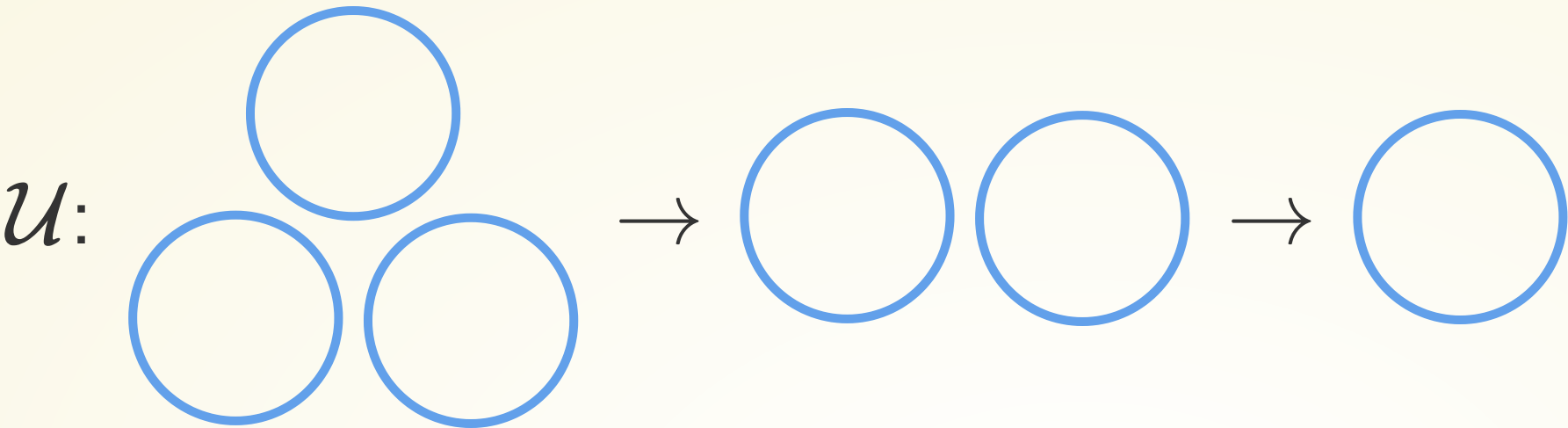


$\mathbb{R}^3 \setminus U_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_3$	$\mathbb{R}^3 \setminus L_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_3$
$\downarrow lcq_2$		$\downarrow lcq_2$		$\downarrow lcq_2$	$\downarrow lcq_2$		$\downarrow lcq_2$		$\downarrow lcq_2$
$\mathbb{Z}^3$	$\longrightarrow$	$\mathbb{Z}$	$\longrightarrow$	0	$\mathbb{Z}$	$\longrightarrow$	0	$\longrightarrow$	0

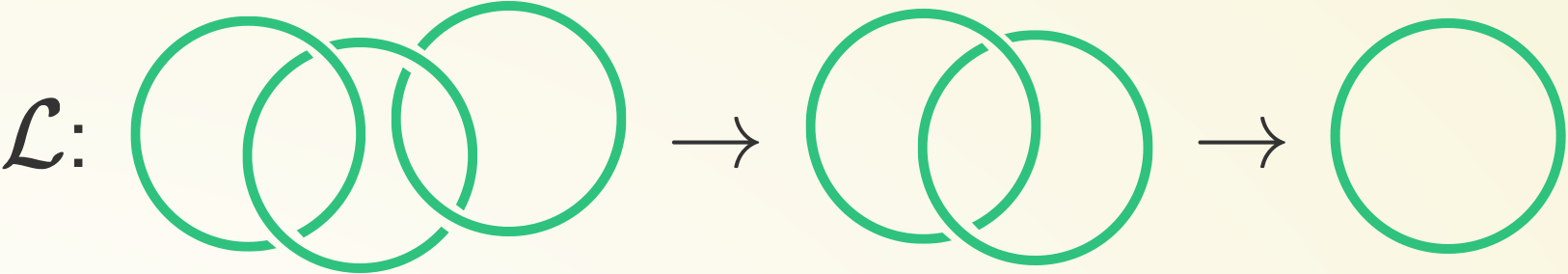
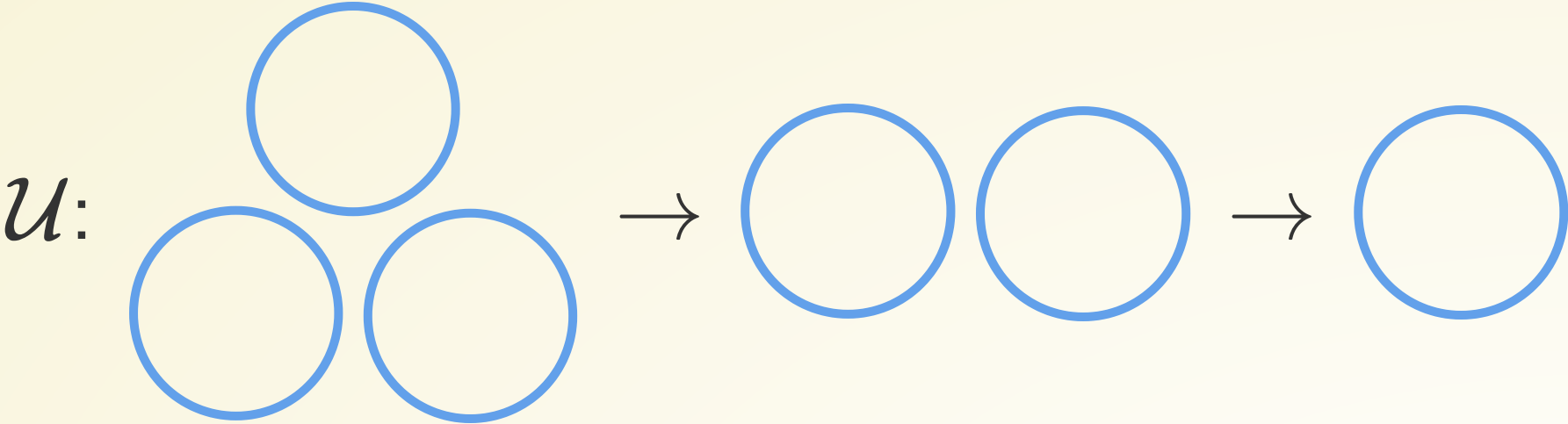


$\mathbb{R}^3 \setminus R_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_3$
$\downarrow lcq_2$		$\downarrow lcq_2$		$\downarrow lcq_2$
$\mathbb{Z}^3$	$\longrightarrow$	$\mathbb{Z}$	$\longrightarrow$	0

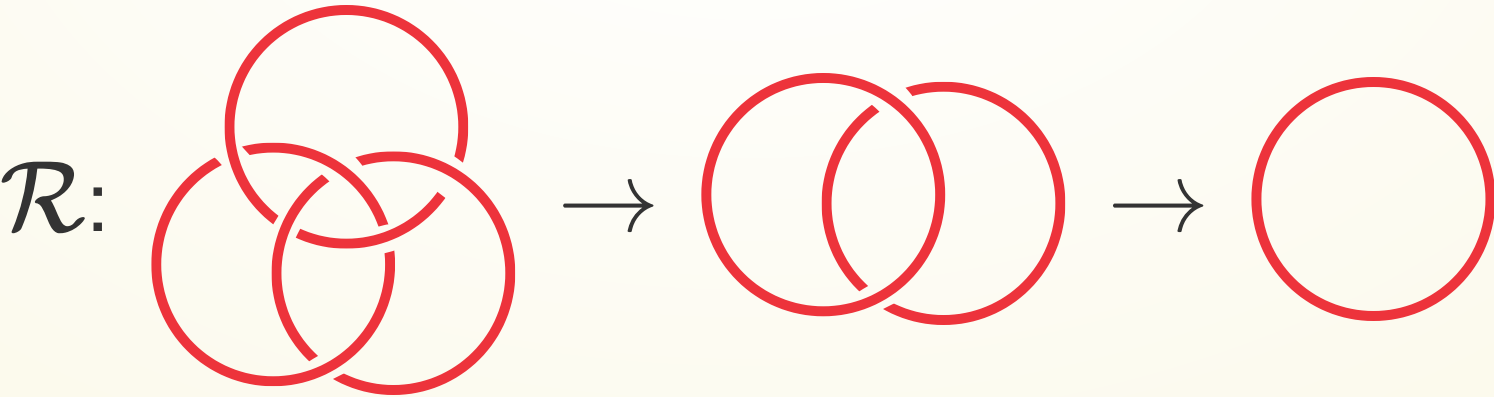
Example: lcq_n barcodes of links. (lcq_2)



Example: lcq_n barcodes of links. (lcq_3)

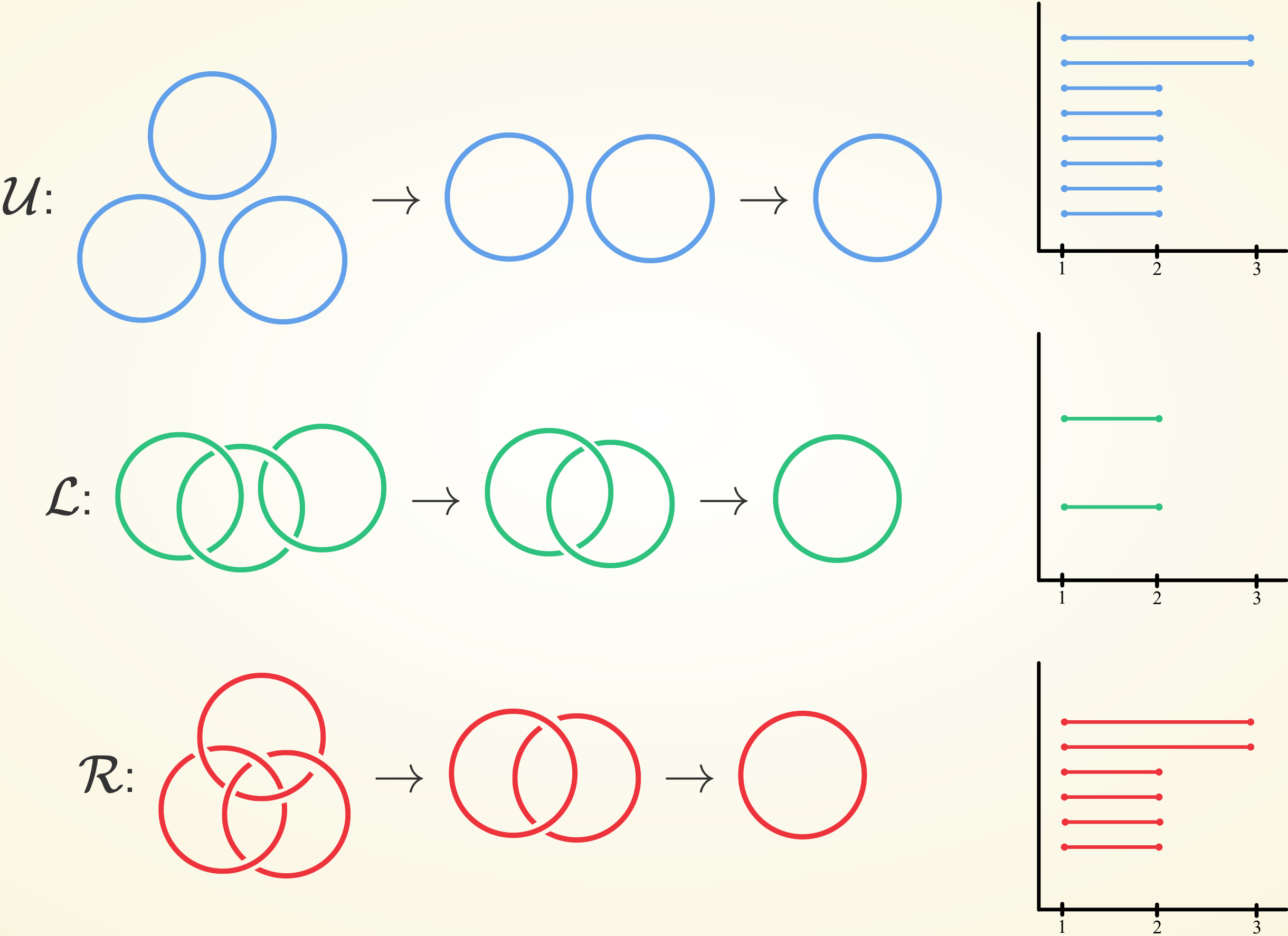


$\mathbb{R}^3 \setminus U_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus U_3$	$\mathbb{R}^3 \setminus L_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus L_3$
$\downarrow lcq_3$		$\downarrow lcq_3$		$\downarrow lcq_3$	$\downarrow lcq_3$		$\downarrow lcq_3$		$\downarrow lcq_3$
$\mathbb{Z}^8$	$\longrightarrow$	$\mathbb{Z}^2$	$\longrightarrow$	0	$\mathbb{Z}^2$	$\longrightarrow$	0	$\longrightarrow$	0



$\mathbb{R}^3 \setminus R_1$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_2$	$\longrightarrow$	$\mathbb{R}^3 \setminus R_3$
$\downarrow lcq_3$		$\downarrow lcq_3$		$\downarrow lcq_3$
$\mathbb{Z}^6$	$\longrightarrow$	$\mathbb{Z}^2$	$\longrightarrow$	0

Example: lcq_n barcodes of links. (lcq_3)



GAP CODE

```
1 LoadPackage("nq");
2
3 LcsQuotient := function(group, i)
4   local h, lcs;
5   nq := NilpotentQuotient(group, i+1);
6   lcs := LowerCentralSeries(nq);
7   if i < Length(lcs) then
8     return AbelianInvariants(lcs[i] / lcs[i+1]);
9   fi;
10  return TrivialGroup();
11 end;
```

Using GAP 4 NQ Package (Nickel, 2007).

ONGOING WORK

- Looking for a computationally efficient algorithm to compute $lcq_n(G)$.
 - Looking for a computationally efficient algorithm to compute a graded Smith normal form.
-

FUTURE WORK

- Apply methods to metric spaces.
- Apply methods to real data.
- Look for other abelian group invariants further than lcq_n .

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- Primož Skraba and Mikael Vejdemo-Johansson. Persistence modules: Algebra and algorithms. *Mathematics of Computation*, 0:0–0, 2013.
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